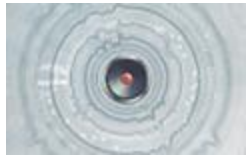
Historical Evolution of Harmonic Qualities in Music: Unravelling Fourier's Tonality, Entropy, and Ambiguity from the 16th to the 20th Centuries

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U.PORTO



DTAMR24

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FCT Fundação para a Ciência e a Tecnologia

…temporal evolution of harmonic qualities…

…prelude…

…with the discrete Fourier transform of pitch-class sets…

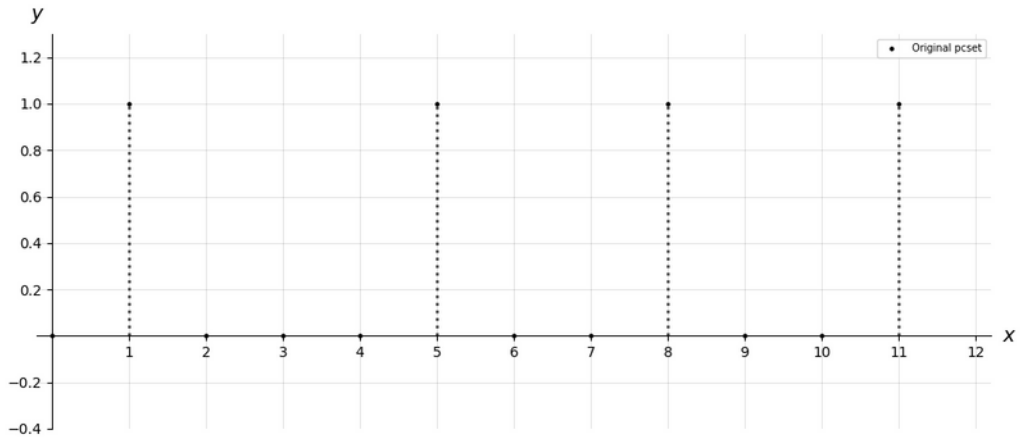
…Act I…

Context and concepts

1. The discrete Fourier transform of pitch-class sets

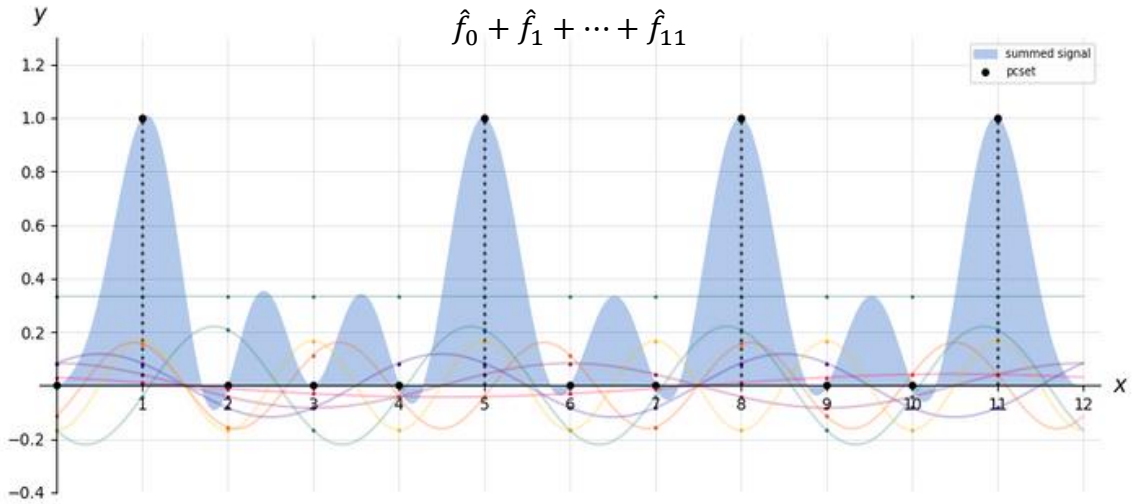
- Signal X is a dominant 7th chord with root in D*b*.

- X = [0, 1, 0, 0, 1, 0, 0, 0, 1, 0, 0, 1]



- *Discrete representation of X*

- The DFT of a pcset [1, 7] produces 6 non-trivial coefficients [9, 10, 14].

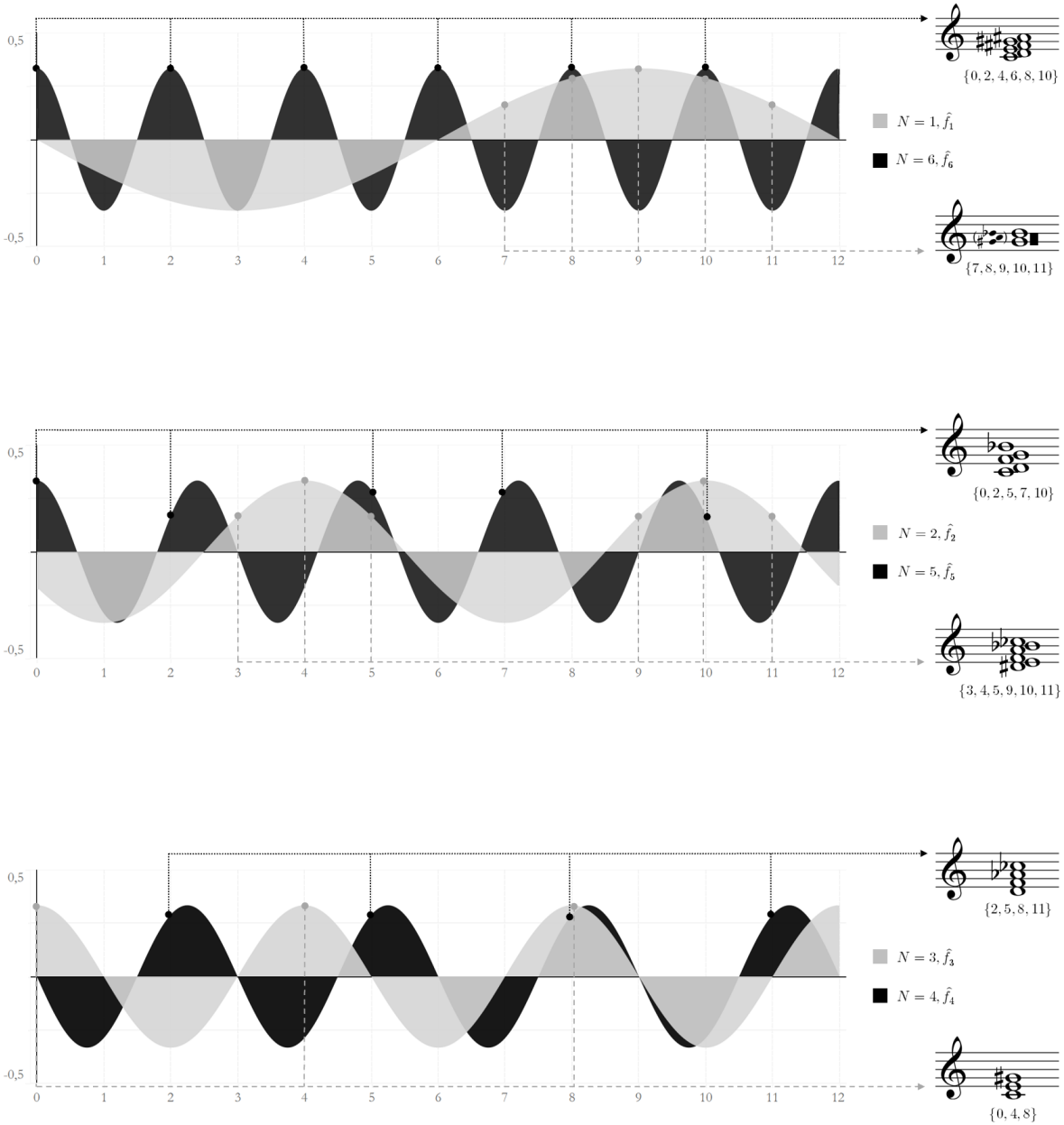


- *DFT coefficients for chord X.*

2. The DFT harmonic qualities

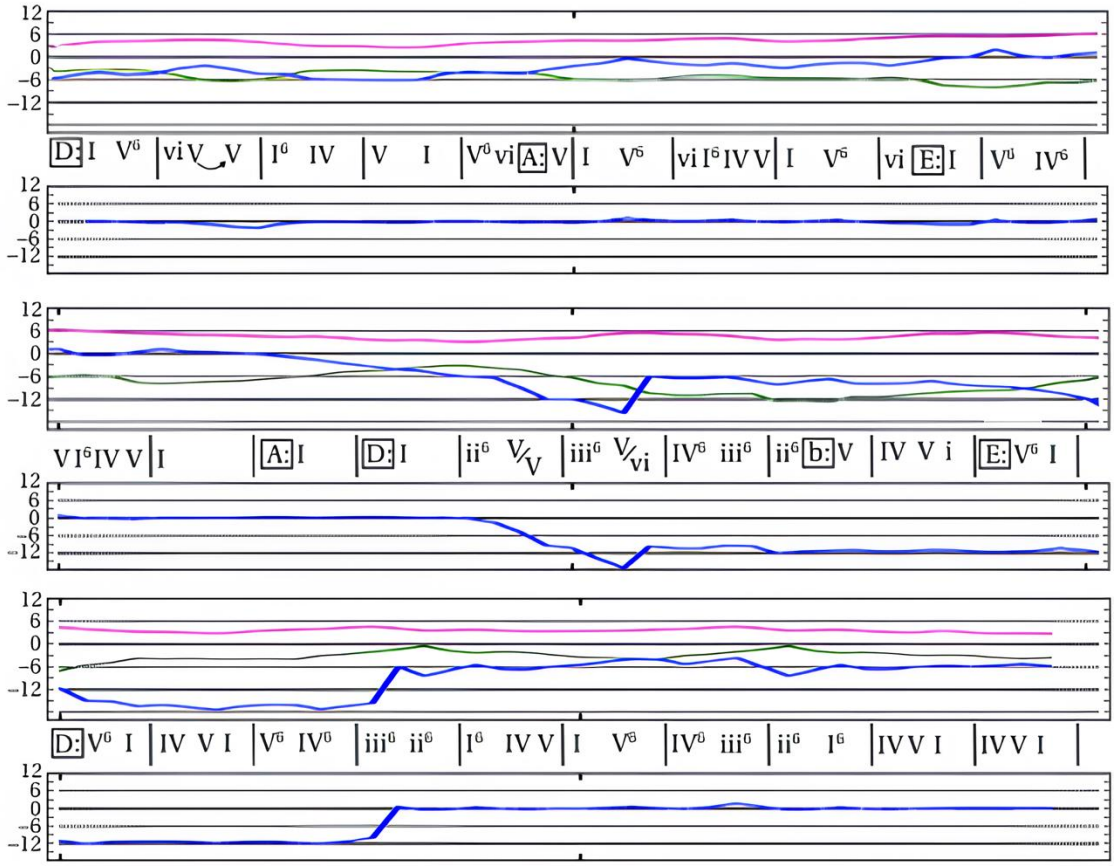
- Association between DFT coefficients and harmonic qualities [1, 3, 10, 13]:

  $|\hat{f}_1|$ - clustered quality;
  $|\hat{f}_2|$ - diadicity;
  $|\hat{f}_3|$ - triadicity;
  $|\hat{f}_4|$ - octatonicity;
  $|\hat{f}_5|$ - diatonicity;
  $|\hat{f}_6|$ - whole-tone quality.



- *DFT coefficients' sinusoids and prototypical chords [6].*

3. The tonal index (2017) [14]



- *DFT Ph2 (green), Ph3 (blue), Ph5 (pink) and the tonal index (blue lower graphs) for Corelli's Violin Sonata Op.5/1 mvt. 2, in Yust (2017). [14]*

- Yust found that:

$$Ph_2 + Ph_3 - Ph_5 \approx 0,$$

i.e., typical tonal distributions keep the quantity *Ph2* + *Ph3* − *Ph5*, the tonal index, close to zero. [14]

4. Entropy of Fourier coefficients (2021) [2]

- Amiot's DFT entropy is given by:

$$H(X) = \sum_{k=1}^{n-1} -p_k \log p_k \ ,$$

being X a pcset with d pitch-classes, n the number of octave divisions, and $|f_k|^2$ the normalised DFT magnitudes. [2]

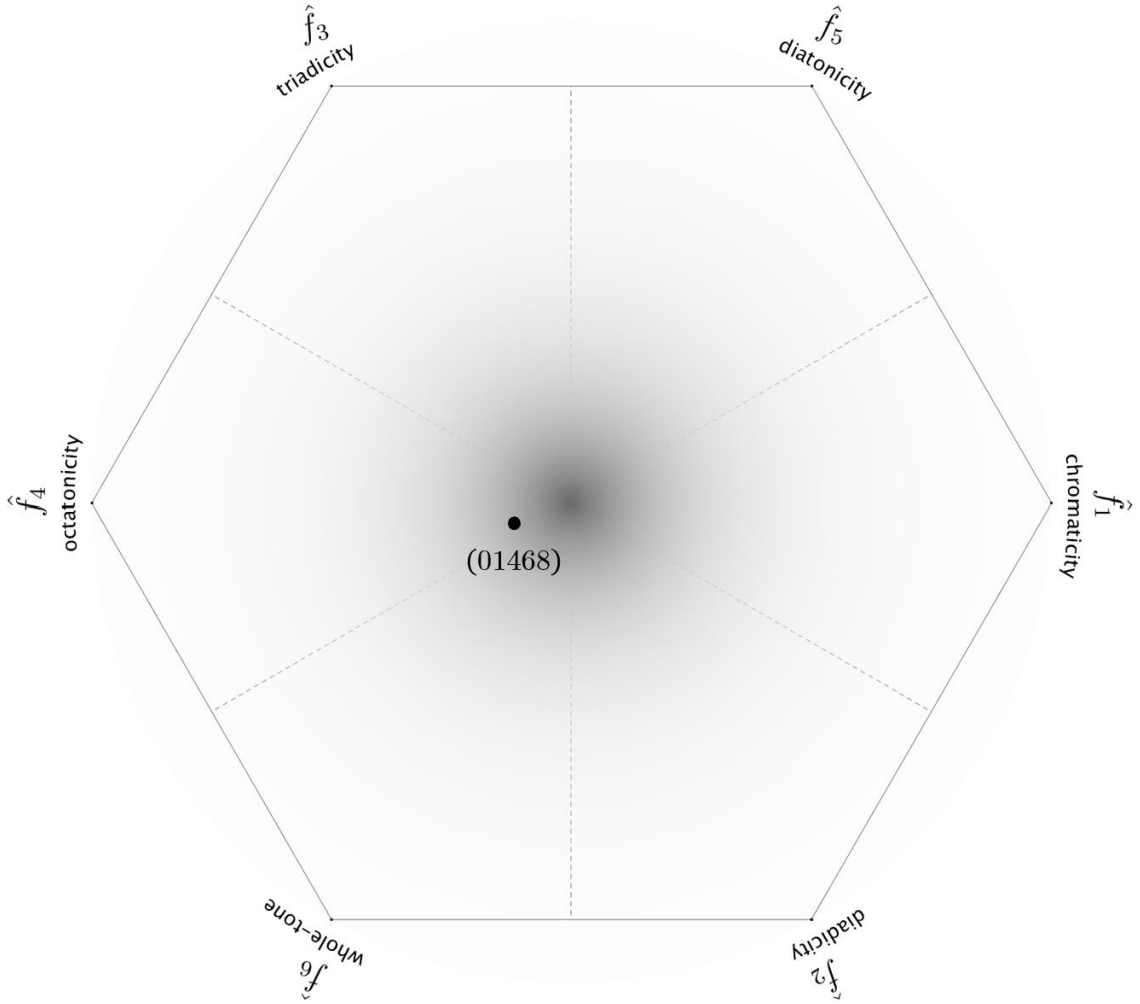
Pc-set	Example	Fourier entropy
Single note	{3}	2.3979
Redoubled note	{5, 5}	2.3979
All-interval chord	{0, 1, 4, 6}	2.3394
Minor seventh	{2, 5, 8, 12}	2.0693
Diminished seventh	{0, 3, 6, 9}	0.6931
Balanced chord	{0, 1, 4, 7, 8}	1.7916
Chromatic chunk	{0, 1, 2, 3, 4}	1.4698
Whole tone	{0, 2, 4, 6, 8, 10}	0.0
Pentatonic	{0, 2, 4, 7, 9}	1.4698
Diatonic	{0, 2, 4, 5, 7, 9, 11}	1.4698
Octatonic	{0, 1, 3, 4, 6, 7, 9, 10}	0.6931
Harmonic Minor	{0, 2, 3, 5, 7, 8, 11}	2.0565
Chromatic - 1	{0,1,2,3,4,5,6,7,8,9,10}	2.3979

- *Fourier entropy for different pcsets. [2]*

5. The Fourier Qualia Space and qualia ambiguity [8]

- In the FQS [8], a set class has 2-dimensional coordinates using the 6-dimensional DFT magnitudes.

- The FQS's centre and its vicinity represent a zone of qualia ambiguity, where two or more harmonic qualities coexist with approximate intensity. [8]



- *Qualia ambiguity in the FQS with the example of set class (01468). [8]*

…Act II…

Diachronic analysis

1. Object of study

- From Yale University, we used the Yale-Classical Archives Corpus.
- https://ycac.yale.edu/

- We extended the YCAC with more than 200 works from composers such as Schoenberg, Stravinsky, Messiaen, Berg or Webern.

Compiling and Processing
The Yale–Classical Archives Corpus
ycac.yale.edu

Christopher Wm. White
University of North Carolina–Greensboro

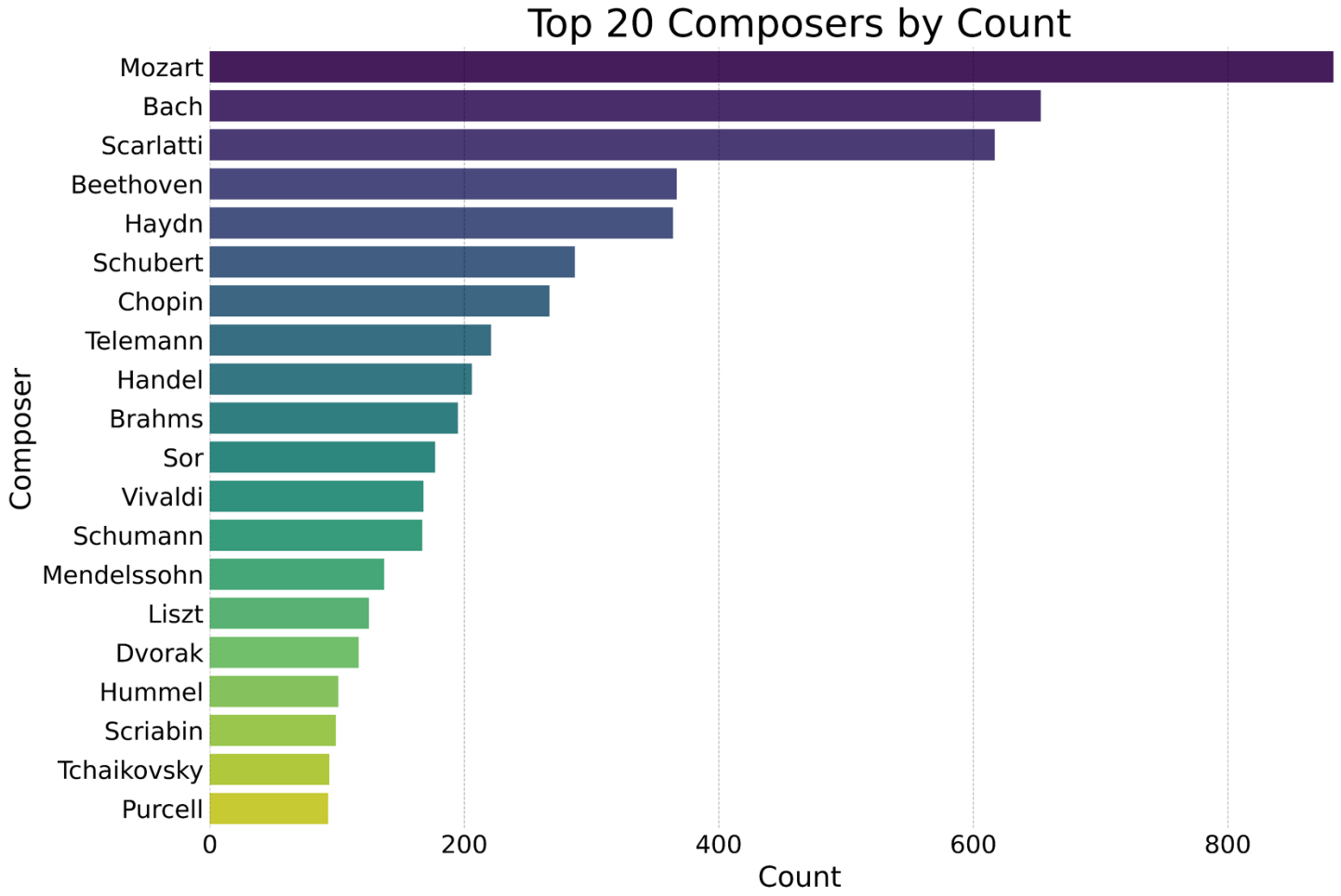
Ian Quinn
Yale University

- *The Yale-Classical Archives Corpus.*

2. Descriptive statistics

a) Composers:

- A total amount of 501 distinct composers;
- Most represented composers are Mozart, Bach, and Scarlatti;
- Diverse collection of musical works, spanning different musical styles and periods.

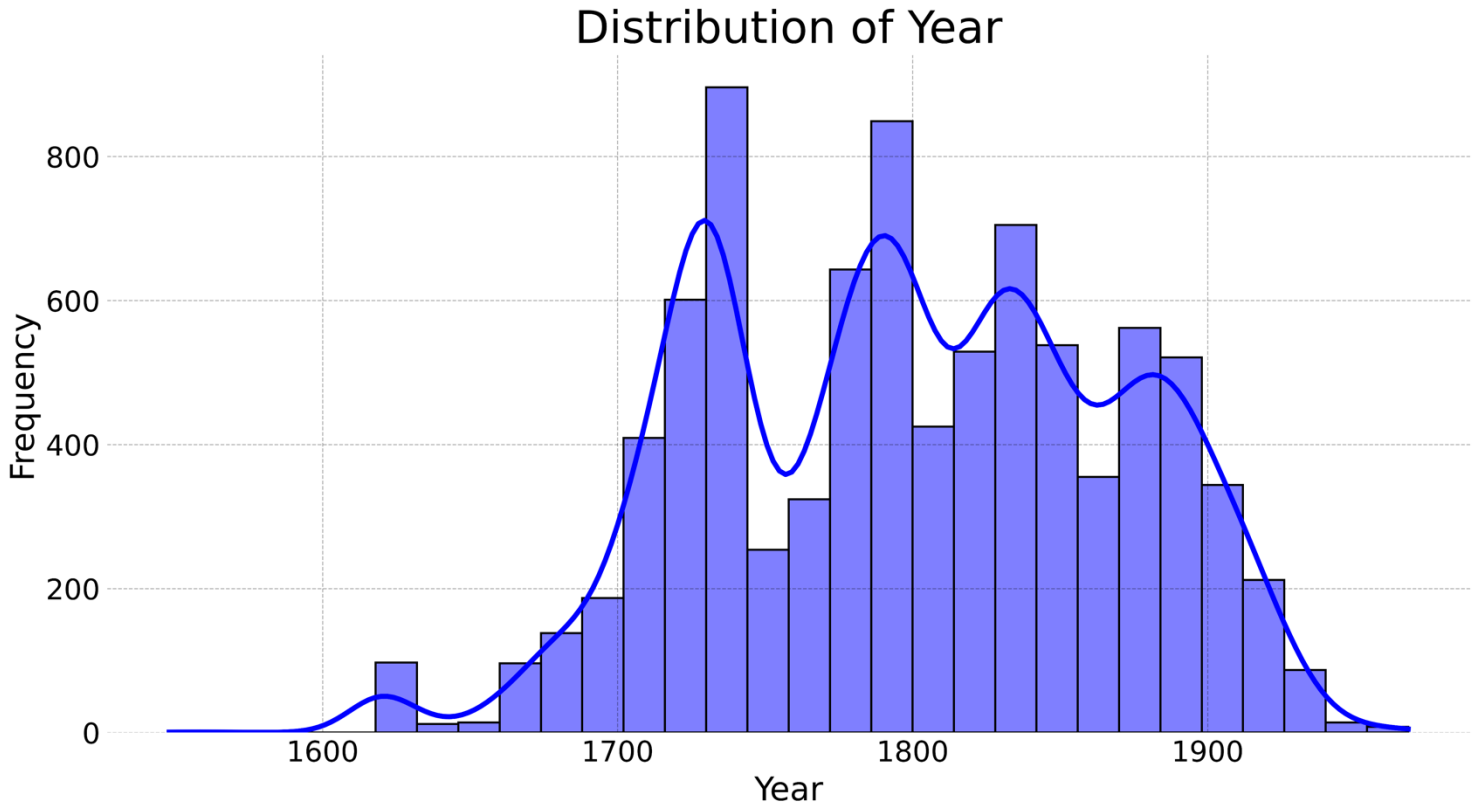


- *Piece count for the 20 most common composers in the dataset.*

2. Descriptive statistics

b) Year distribution:

- Slightly right-skewed distribution;
- Late-baroque, classical, and early-romantic periods are predominant;
- Pre-1700 and post-1900 works are less prominent.



- *Piece count over the past 400 years in the object of study.*

2. Descriptive statistics

	MEAN	MEDIAN	STD
Year	1799.289	1798	68.63795
Cardinality	2.836251	2.823274	0.466415
|F1|	0.938971	0.932784	0.145281
|F2|	1.048271	1.042579	0.117129
|F3|	1.590086	1.593433	0.172652
|F4|	1.572863	1.573806	0.187636
|F5|	1.670702	1.663617	0.196269
|F6|	1.072041	1.068348	0.156022
Ti	-0.01614	-0.0568	1.080621
H	2.043134	2.050232	0.035881
Amb	0.733773	0.735664	0.020757

- *Descriptive statistics (mean, median, and standard deviation) of the computed variables.*

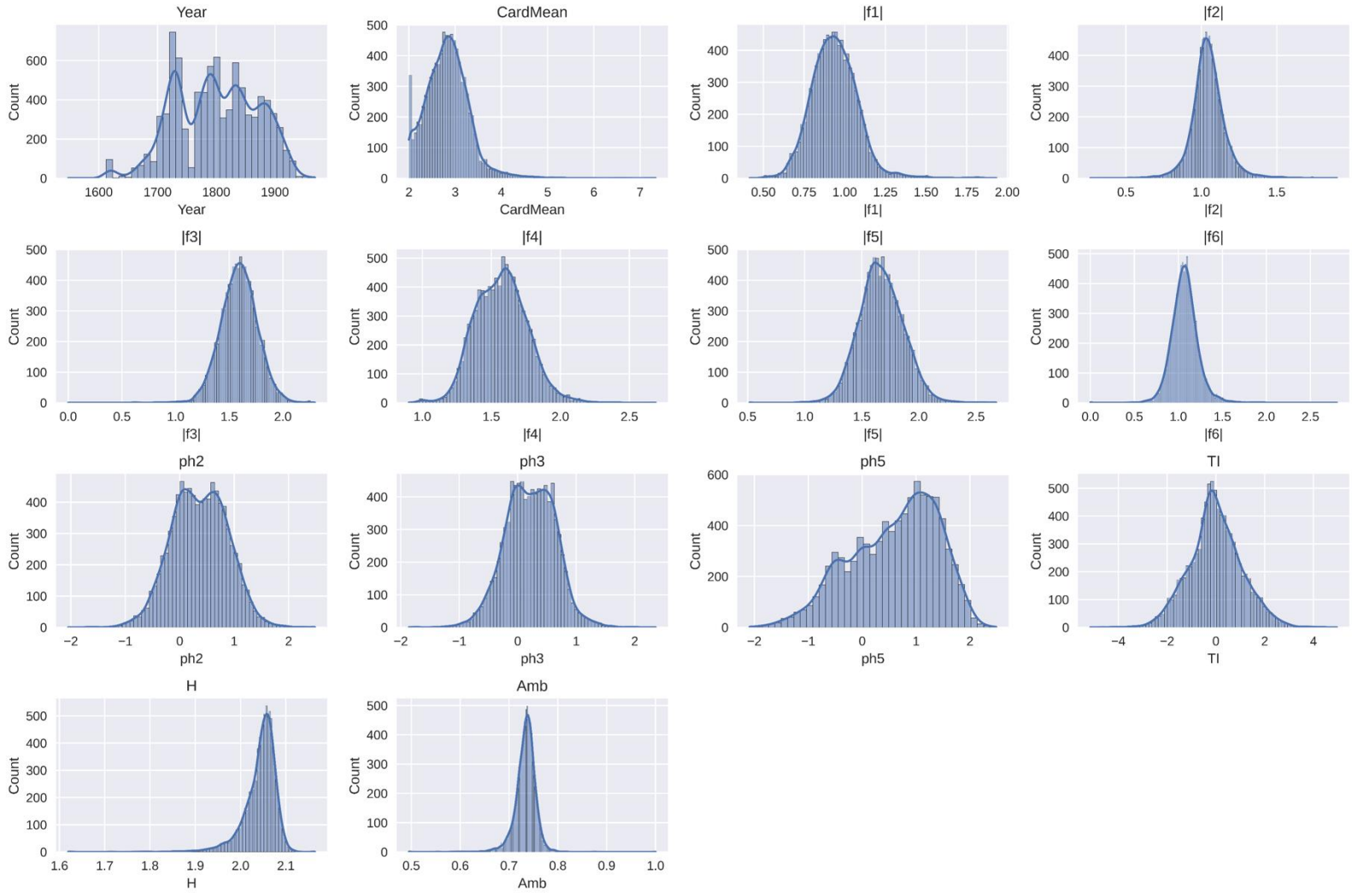
c) Variables computed in our analysis:

- Year: when the work was composed;
- CardMean: the mean cardinality observed in the work;
- $|f_{1,\ldots,6}|$: the magnitude of the non-trivial DFT coefficients 1 to 6;
- TI: the tonal index;
- H: the entropy of the DFT coefficients;
- Amb: qualia ambiguity.

2. Descriptive statistics

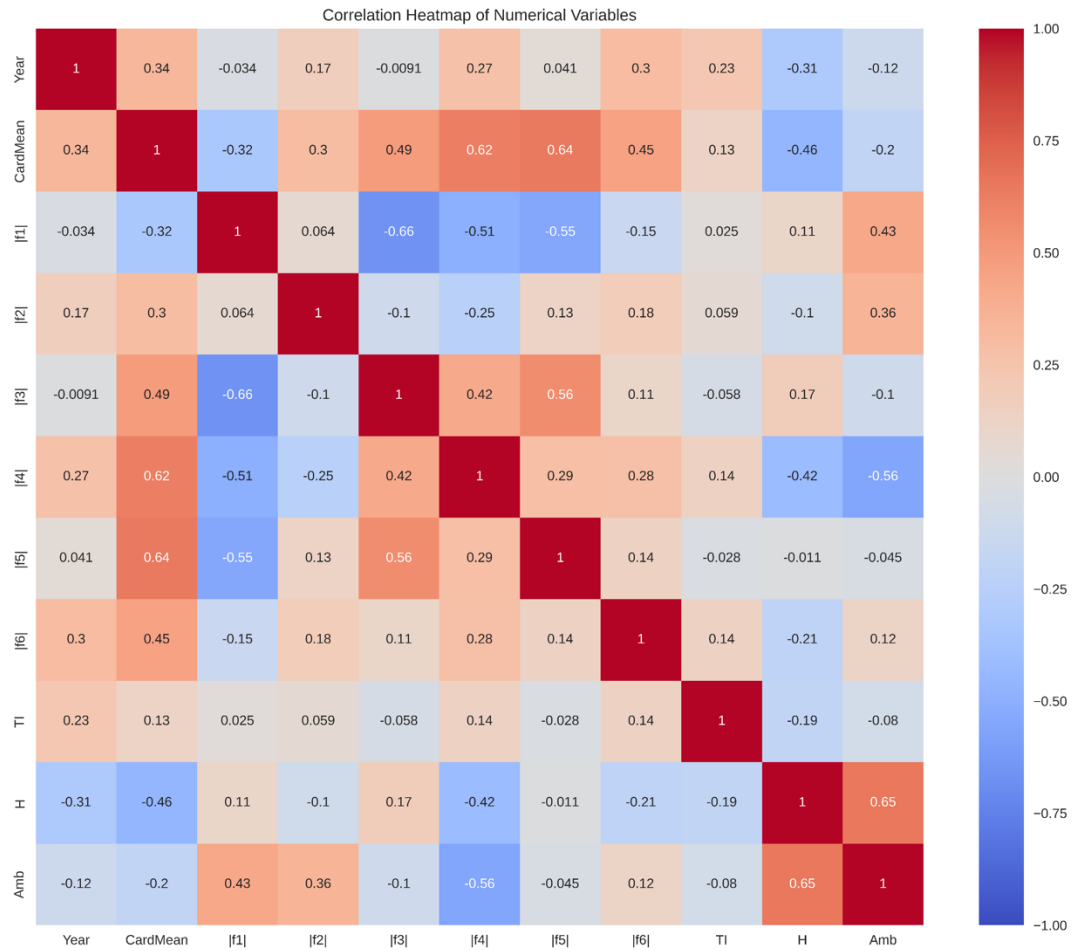
d) Remaining distributions:

- Mean cardinality highly left-skewed around the number 3;
- Tonal index close to a normal distribution around 0, since the corpus is mainly tonal.



- *Distribution plots of the computed variables.*

3. Correlation matrix



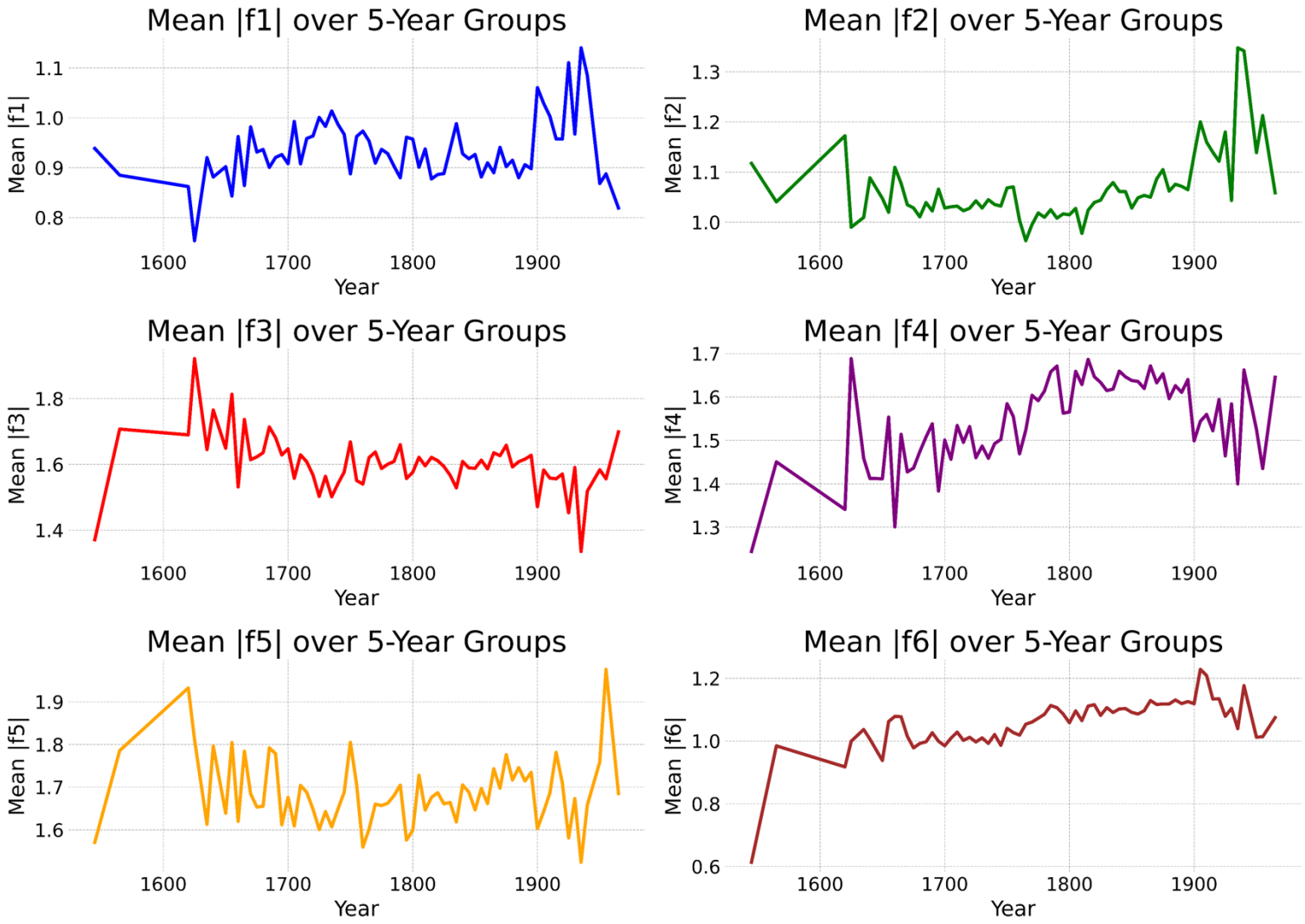
- *Correlation heatmap between all computed variables. Red colour indicates high positive correlation and blue colour high negative correlation.*

- Strong positive correlations: entropy with ambiguity (.65); cardinality with $|f_5|$ (.64), and cardinality with $|f_4|$ (.62);
- Strong negative correlations: $|f_1|$ with $|f_3|$ (-.66);
- Moderate negative correlation: $|f_5|$ with $|f_1|$ (-.55), and cardinality with entropy.

4. Time-series graphs

a) Individual DFT magnitudes:

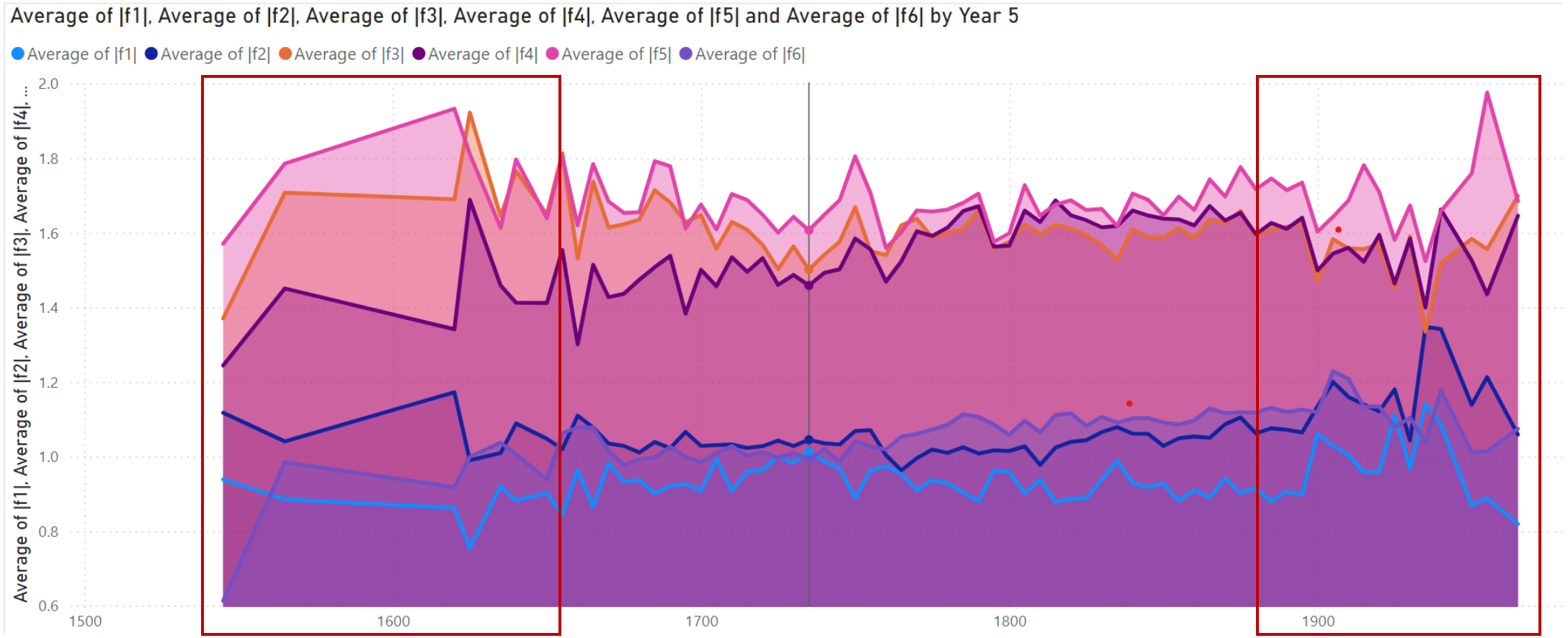
- Clustered quality ($|f_1|$) and diadicity ($|f_2|$) surges in the 20th century;
- Stable levels of triadicity ($|f_3|$) and diatonicity ($|f_5|$);
- Increasing values of octatonicity ($|f_4|$) and whole-tone ($|f_6|$) over time.



- *Single time-graphs for the 6 DFT coefficients.*

4. Time-series graphs

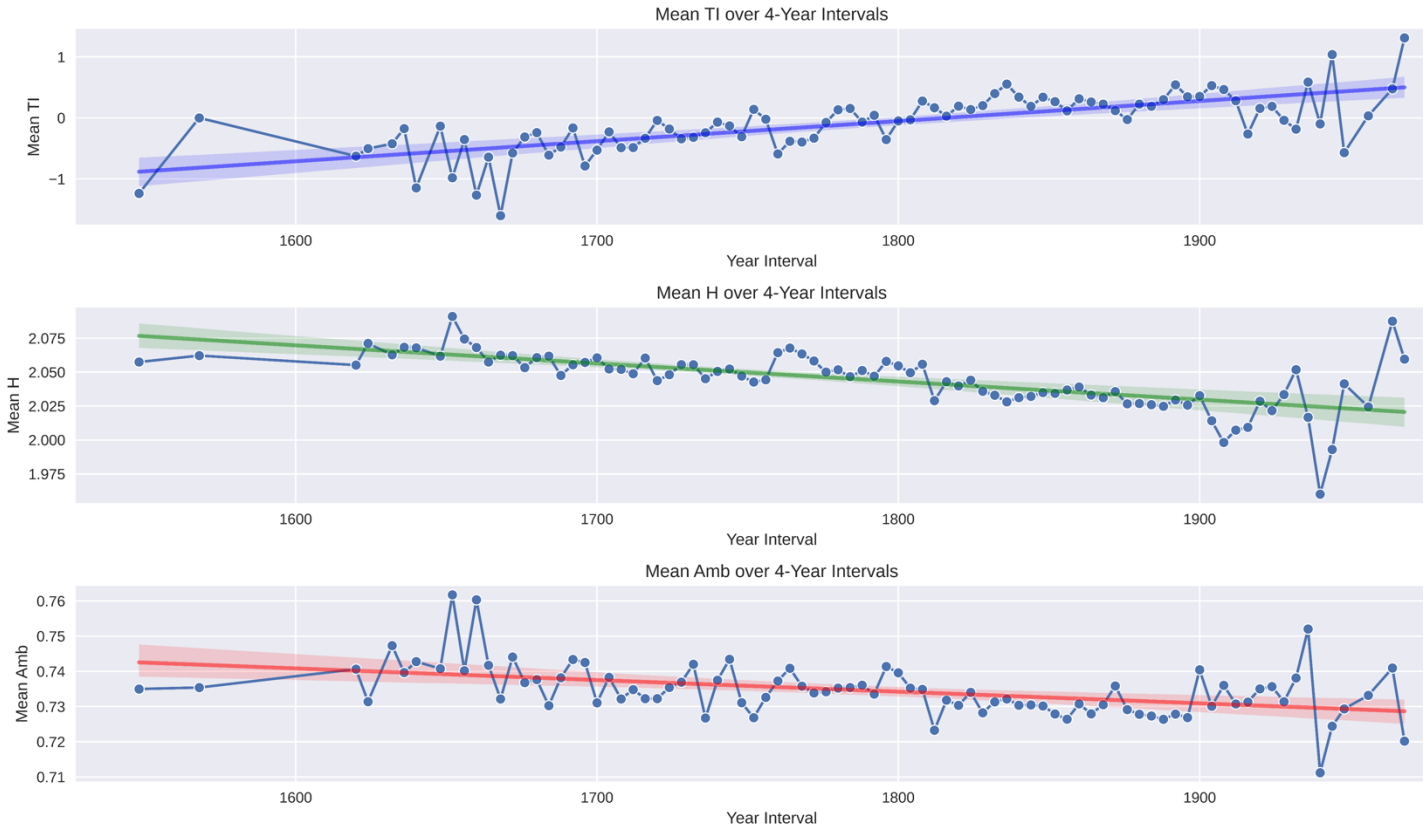
b) Combined DFT magnitudes plot



- *Combined plot of the 6 non-trivial DFT coefficients over time. Color scheme:*

$|f_1|$, $|f_2|$, $|f_3|$, $|f_4|$, $|f_5|$, $|f_6|$.

4. Time-series graphs



- *On the top, historical evolution of the tonal index, on the middle graph the evolution of entropy, and on the bottom ambiguity trend over time.*

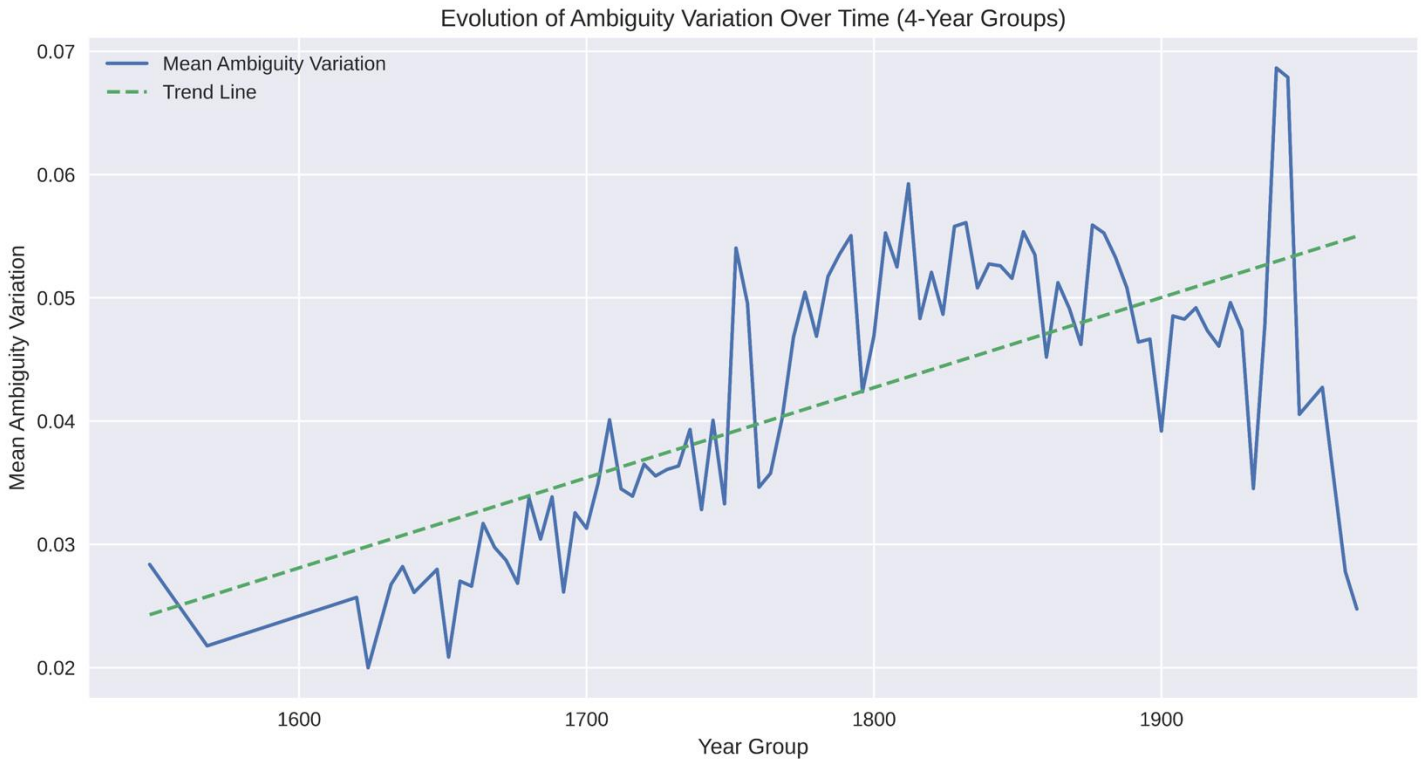
c) TI, H, and Ambiguity in comparison:

- Visualisation of the negative and positive correlations;
- TI close to 0 during the tonal functional period and > 0 towards the 20th century;
- Tendency for more organized pitch structures (lower H) over time;
- Tendency for a slight decrease in qualia ambiguity?

4. Time-series graphs

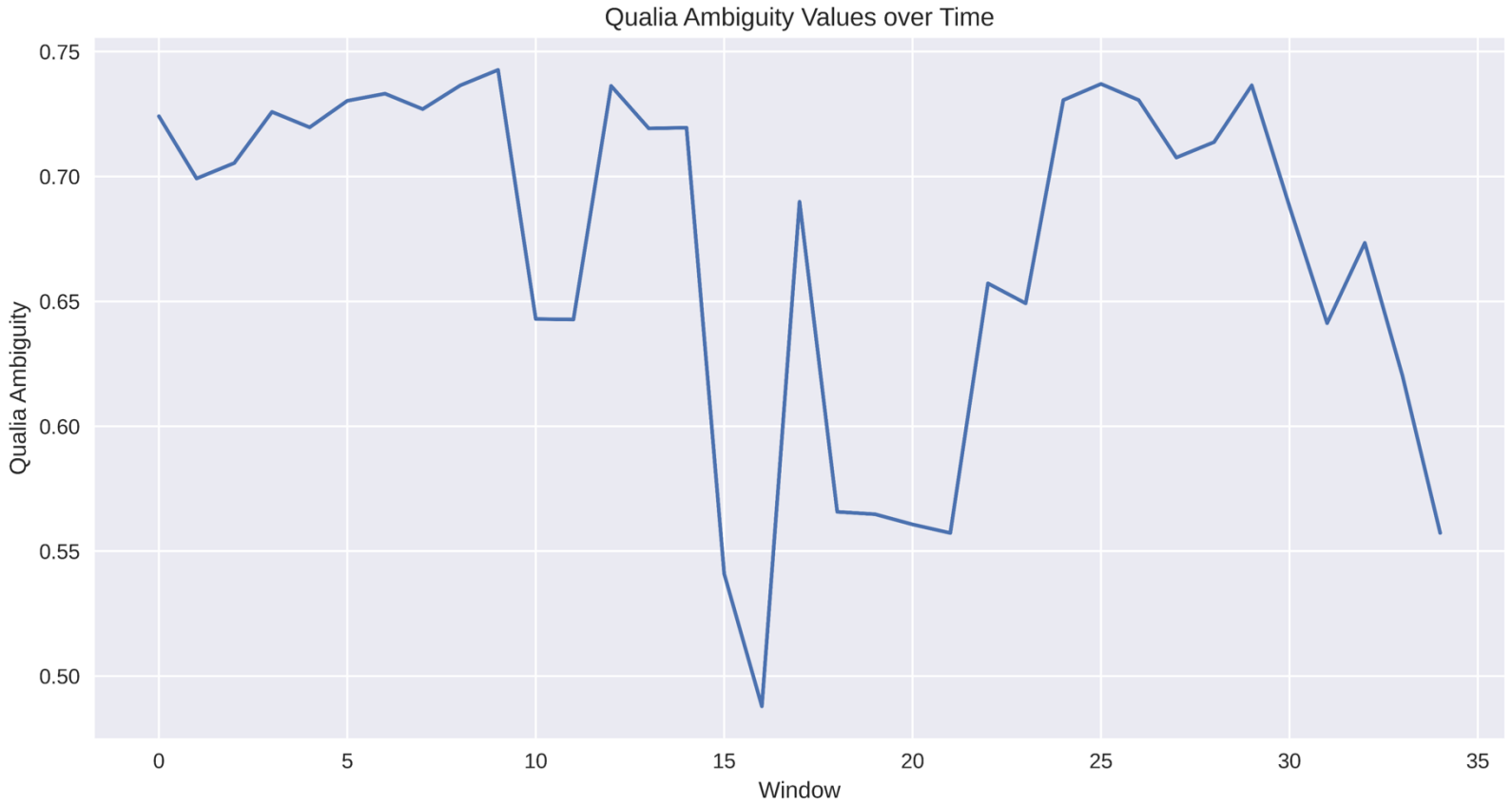
d) Ambiguity's standard deviation:

- Assessment of how much ambiguity variation a piece contains;
- Tendency for progressively higher ambiguity variation towards the 20th century.



- *The standard deviation per year of ambiguity values registered in a windowed analysis of the corpus. Regression line in dashed green.*

5. The curious case of Purcell's *Cold Song*



- High ambiguity variation in late 16th century

?

- *Purcell's Cold Song ("What Power Art Thou", King Arthur Opera, 1691) windowed analysis of qualia ambiguity values.*

…interlude…



- Painting of a *Thames frost fair,* London.

What power art thou, who from below

Hast made me rise unwillingly and slow

From beds of everlasting snow?

See'st thou not how stiff and wondrous old,

Far unfit to bear the bitter cold,

I can scarcely move or draw my breath?

Let me, let me freeze again to death.

John Dryden

…Act III…

Synchronic analysis

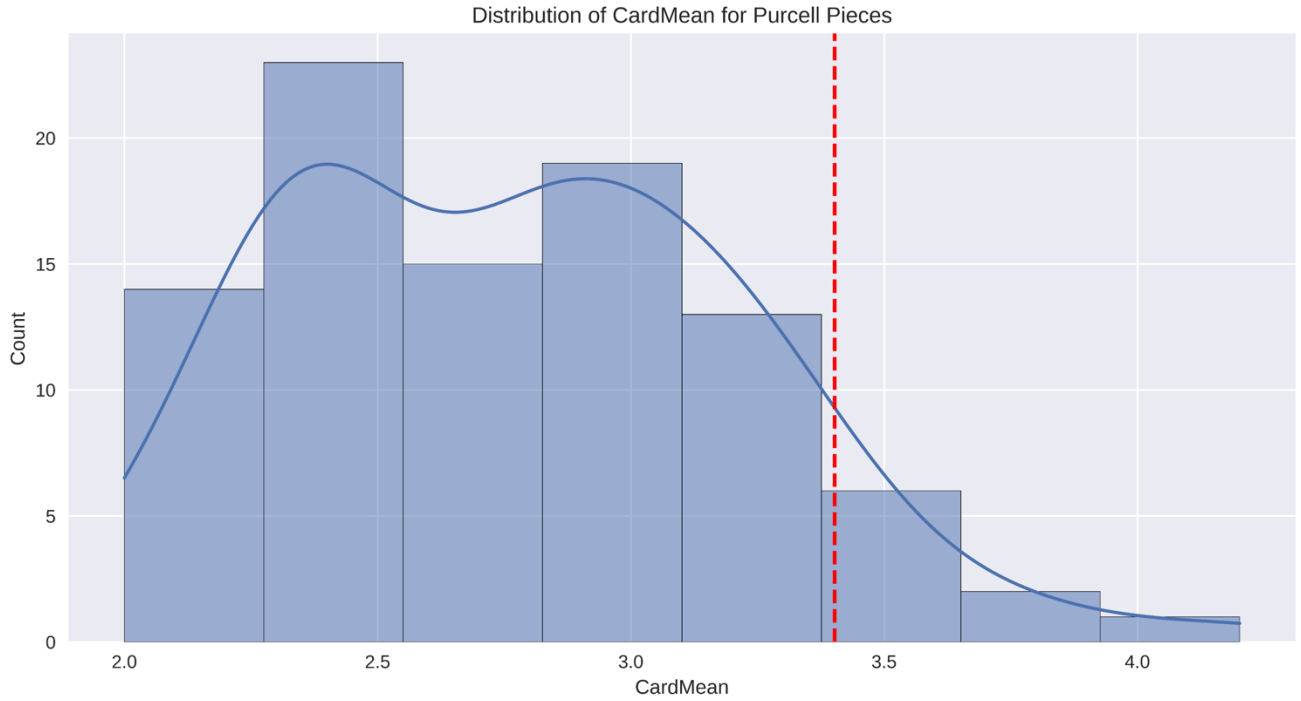
1. First remarks on the *Cold Song*

- Act III of Purcell's *King Arthur* opera.
- Repeated notes in the vocal line suggest chattering teeth.
- Descending or ascending melodic lines resemble sinking or rising into/from cold.
- Chromaticism and dissonances create a sense of discomfort associated with cold.



- *Repetitive texture of Purcell's Cold Song, as a resonator of cold and freeze.*

2. *Cold Song*'s mean cardinality

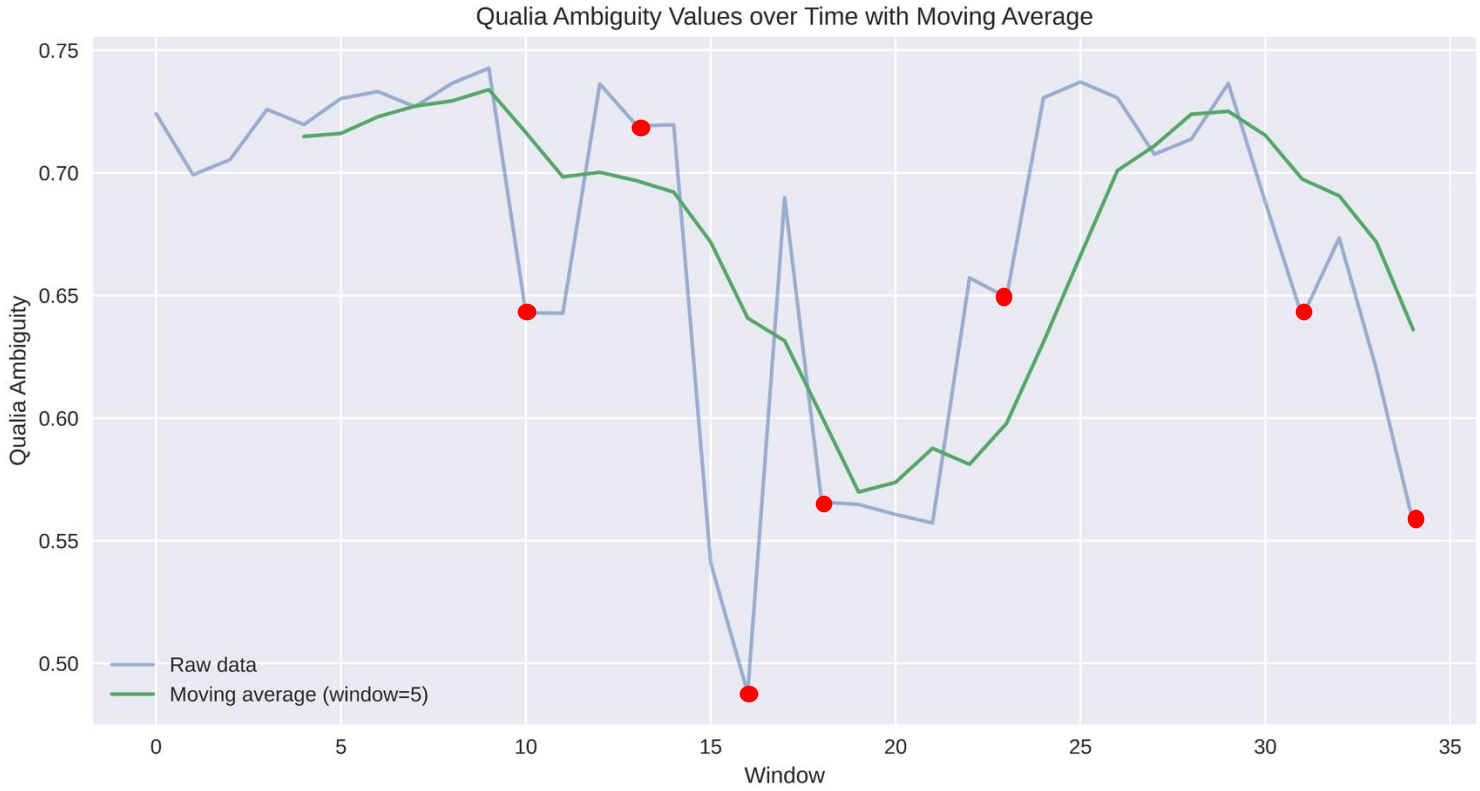


- *Distribution of the mean cardinality in Purcell's works. The red dashed line stands for the mean cardinality in his Cold Song.*

- Mean cardinality in Purcell's *Cold Song* is 3.4.
- The piece has high mean cardinality, when compared with other Purcell's pieces in the dataset.

3. Ambiguity graph, once again

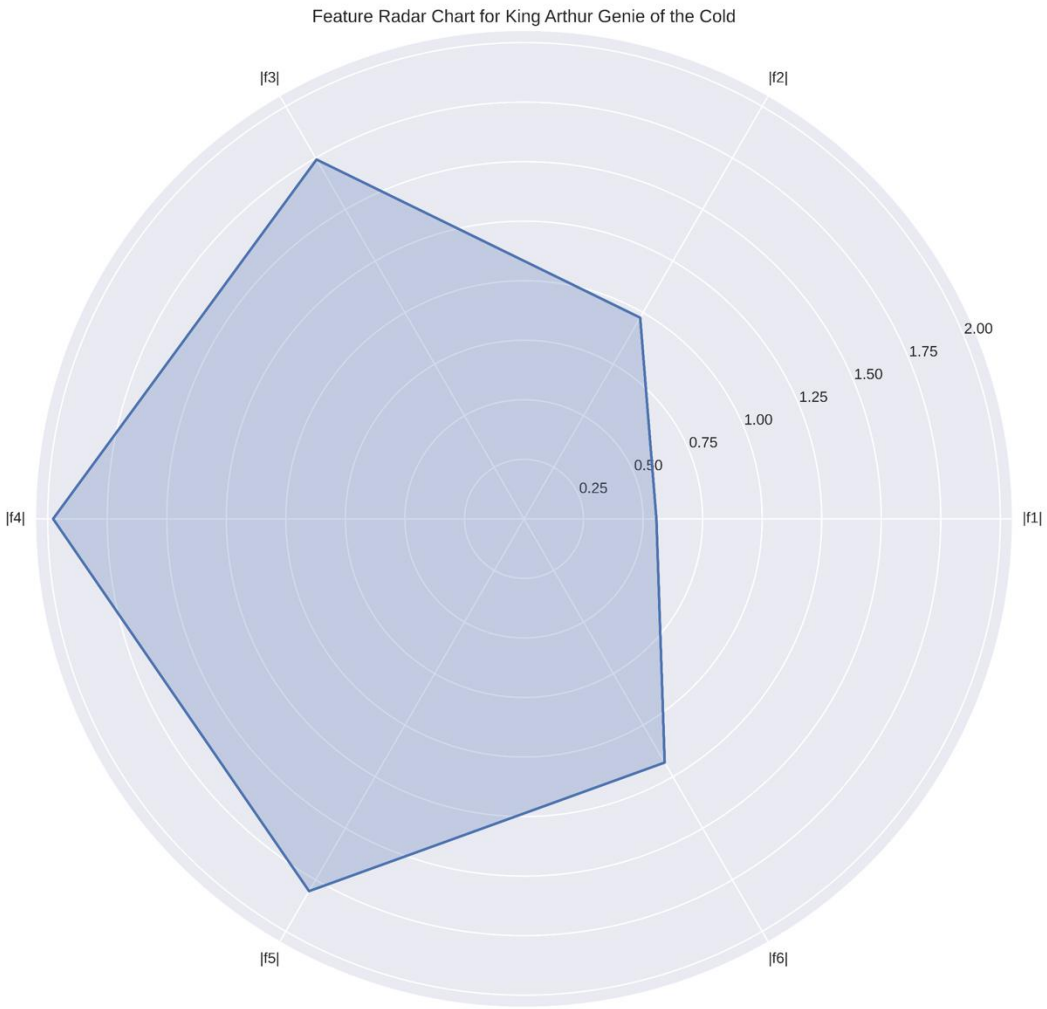
- A striking feature in the qualia ambiguity graph is the fact that sudden drops in value coincide with various phrase cadences.
- Cadences in bars 10, 13, 16, 18, 23, 31, and 33 align with drops in qualia ambiguity.



- *Purcell's Cold Song windowed analysis of qualia ambiguity values. In green moving average for tendency line. Red dots represent musical cadences. Window = bar.*

4. DFT magnitude profile of the *Cold Song*

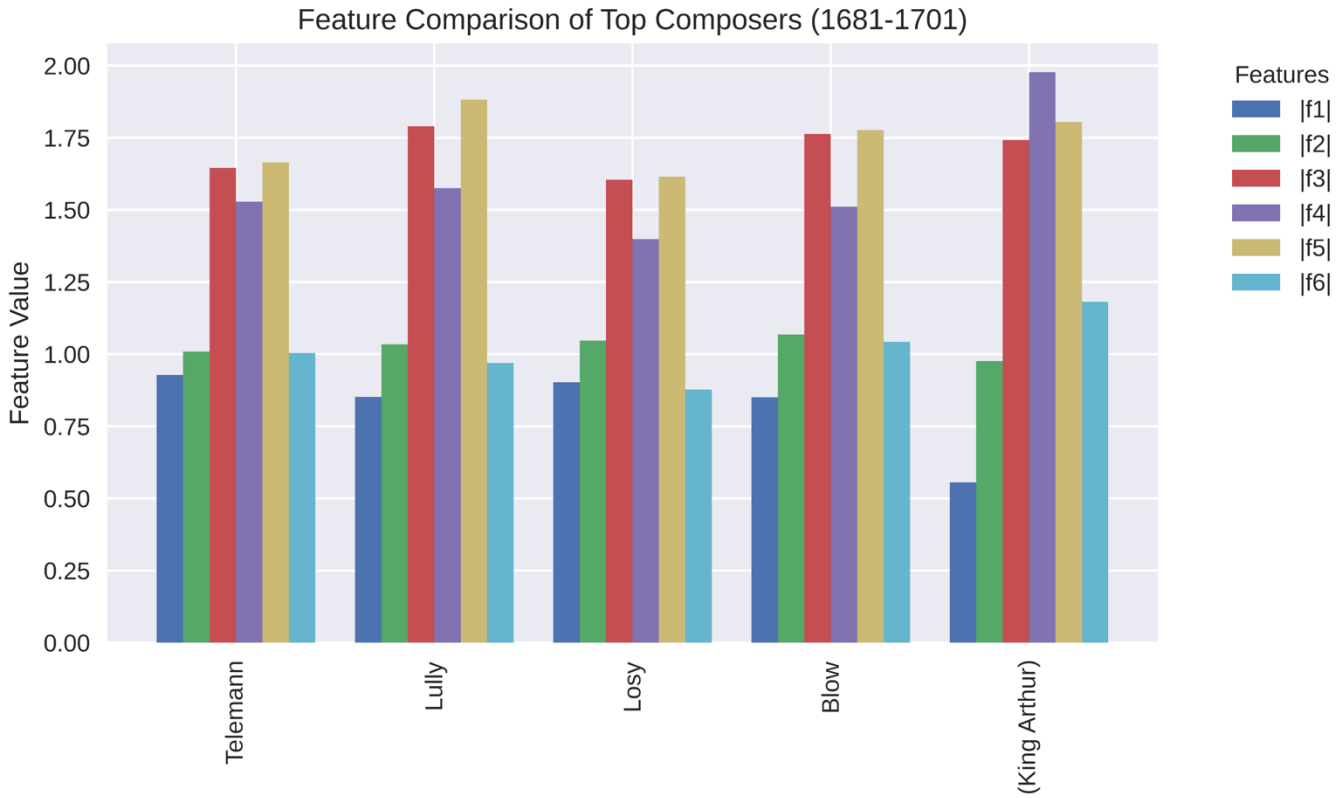
- The unexpected feature of the chart is the prominence of $|f_4|$ (octatonicity).
- Prominent $|f_3|$ (triadicity) and $|f_5|$ (diatonicity) are also noteworthy but not as unexpected.



- *Radar chart comparing the 6 non-trivial DFT coefficient magnitudes in the Cold Song.*

5. Viewing the *Cold Song* in its context

- Prominent values of $|f_4|$ are confirmed.
- Is this piece octatonic?
- Are there emerging octatonic structures in the cold song?

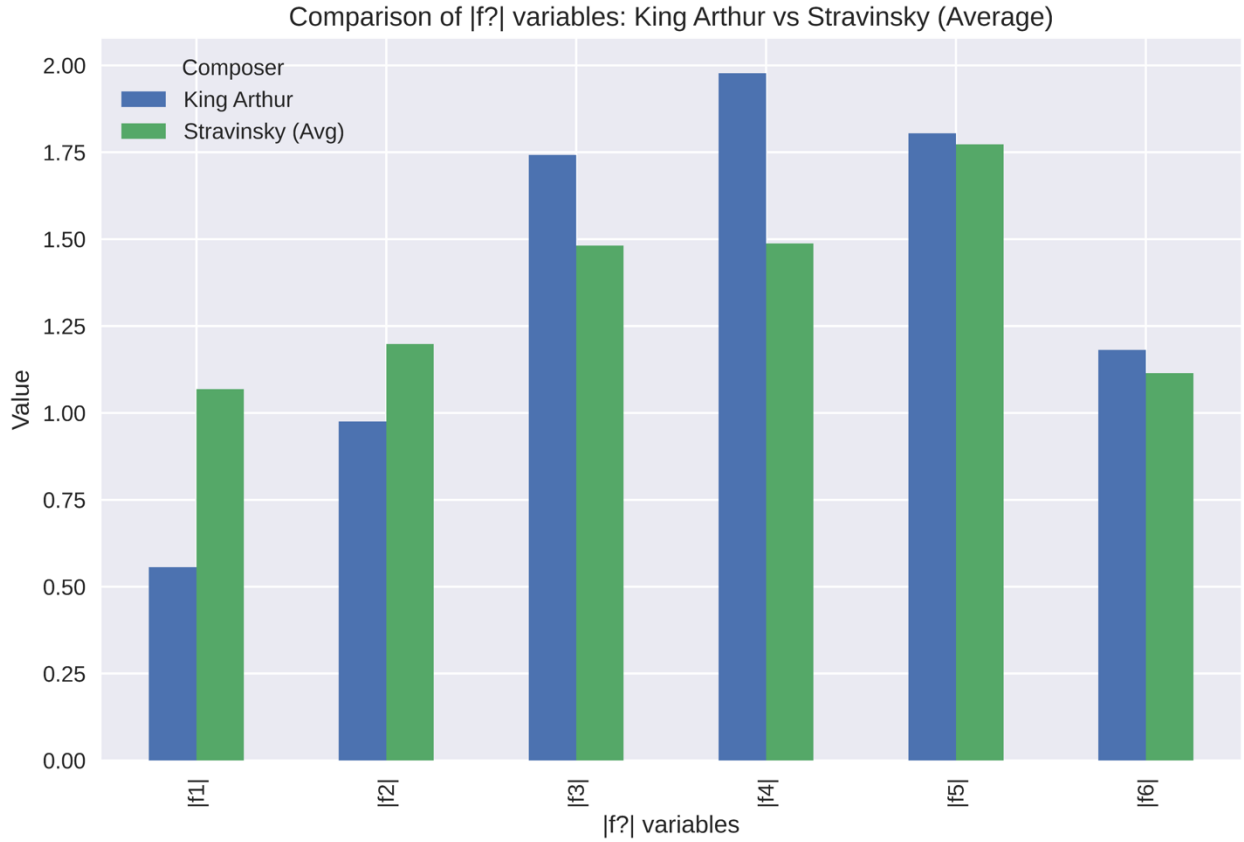


- *Barplot comparing the 6 non-trivial DFT coefficient magnitudes between Purcell's Cold Song and 4 contemporary composers.*

6. What's happening with octatonicity in the *Cold Song*?

- Higher octatonicity ($|f_4|$) values than Stravinsky's works

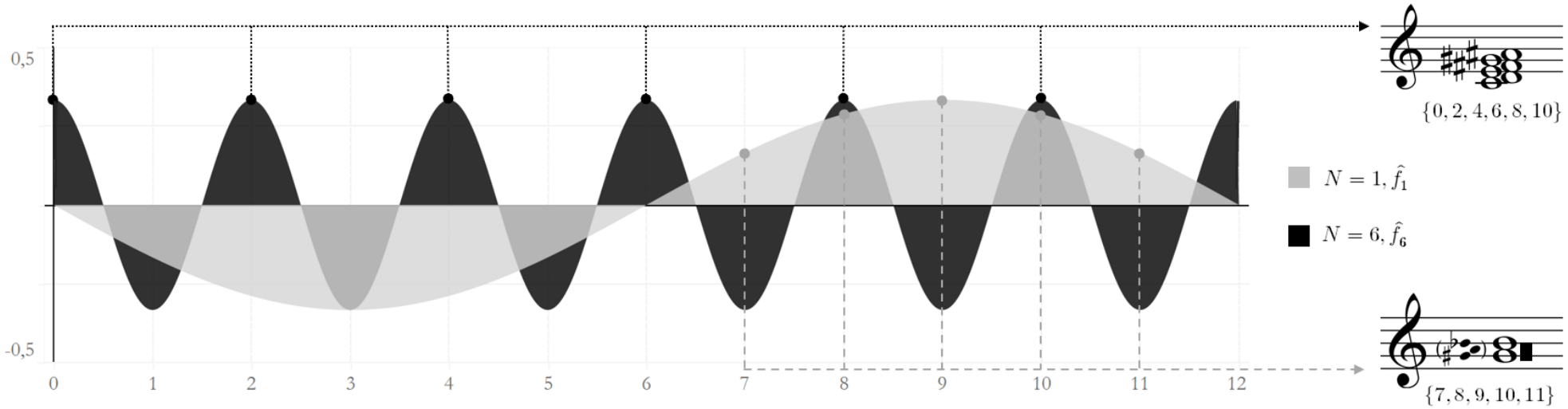
?



- *Barplot comparing the 6 non-trivial DFT coefficient magnitudes between Purcell's Cold Song and Stravinsky's works.*

7. A nomenclature problem

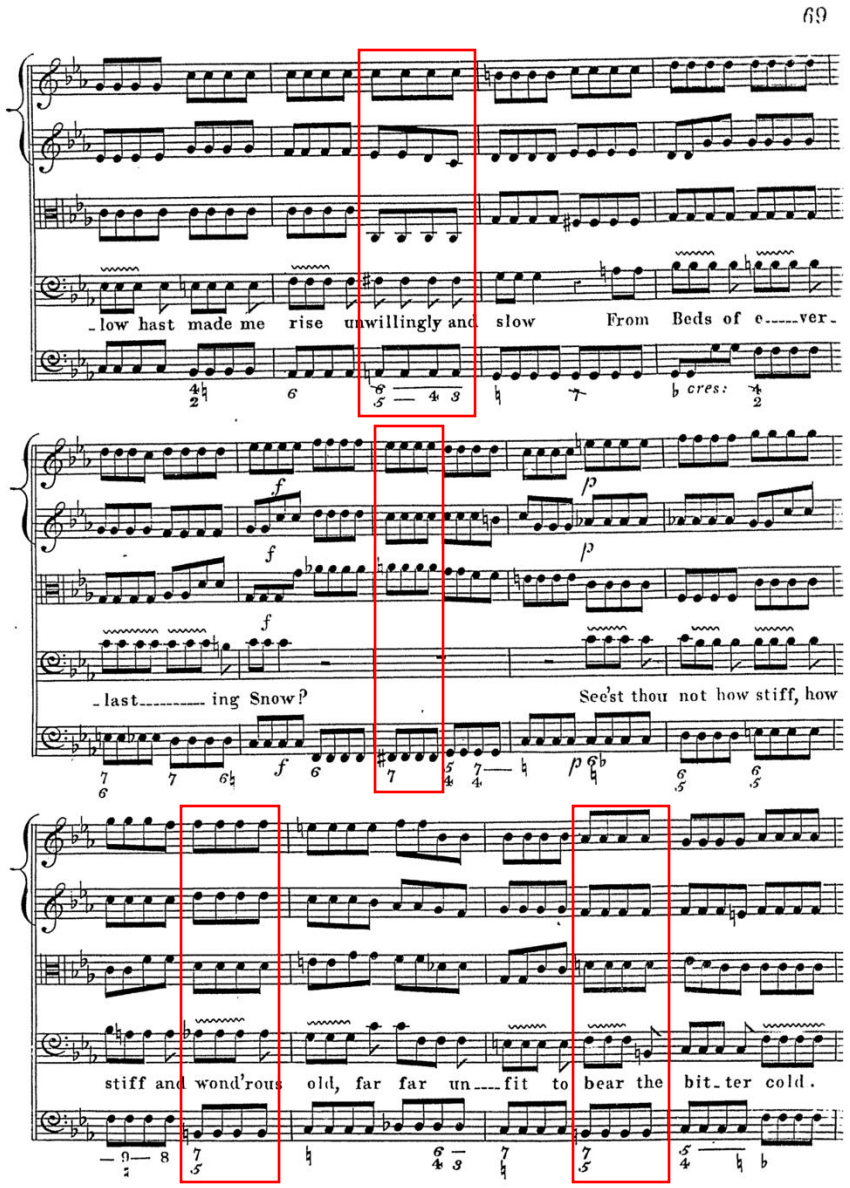
- According to its periodicity and phase value, the same DFT coefficients can resonate with slightly different harmonic qualities;
- Hence, $|f_4|$ can be interpreted as a measure for octatonicity or diminished quality. [6]



- *Depiction of two DFT coefficients represented as sinusoids and the respective harmonic qualities and prototypes that resonate given their periodicity and phase.*

8. The diminished quality in the *Cold Song*

- The magnitude in $|f_4|$ can be a manifestation minor thirds, tritones, diminished 7th chords, or even octatonicity.
- Indeed, the diminished 7th and the tritone do play a structural role in Purcell's work.
- A generalised nomenclature for the DFT harmonic qualities would be useful.

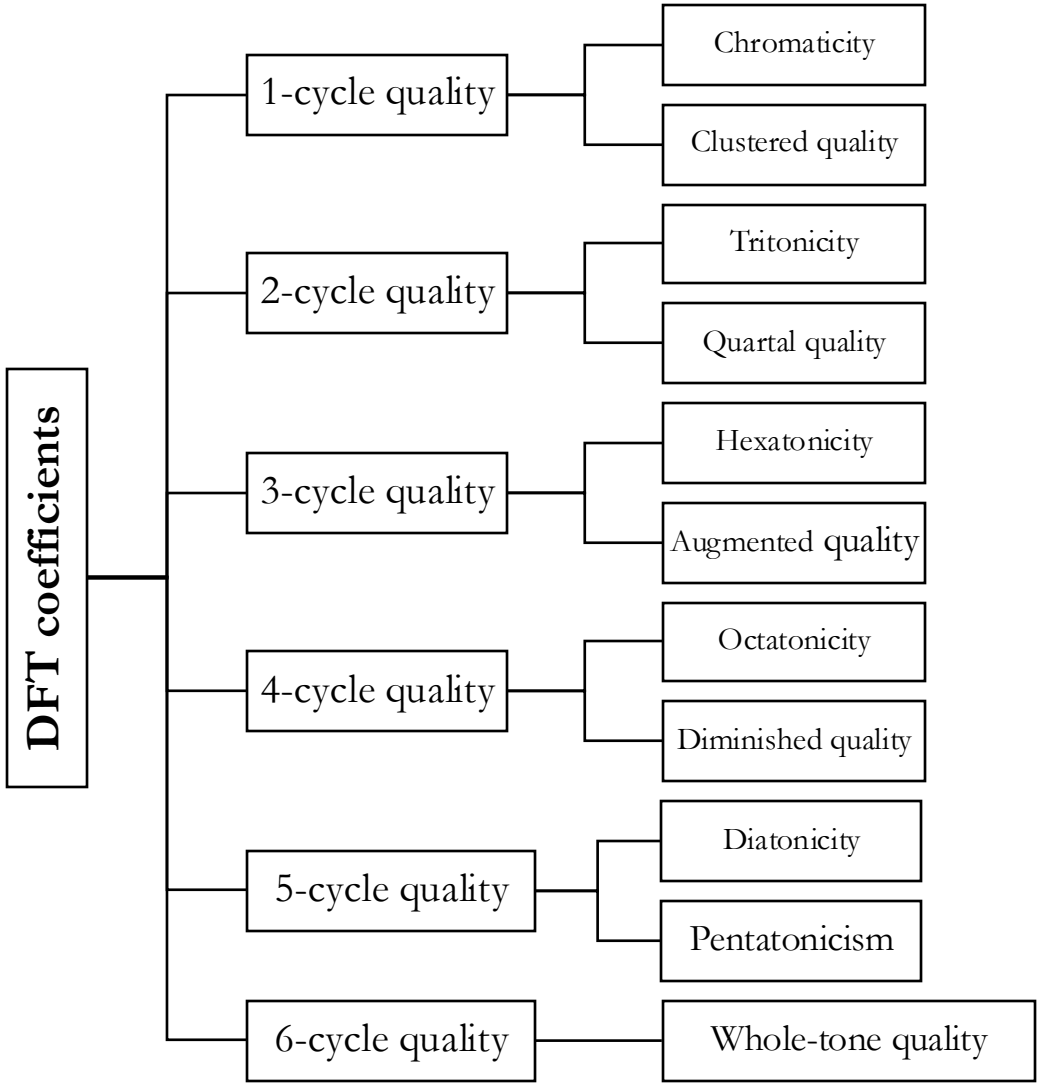


- *Manifestations of the diminished quality in Purcell's Cold Song.*

9. One possible solution

- Association between DFT coefficients and harmonic qualities:

$|\hat{f}_1|$ - 1-cycle quality;
$|\hat{f}_2|$ - 2-cycle quality;
$|\hat{f}_3|$ - 3-cycle quality;
$|\hat{f}_4|$ - 4-cycle quality;
$|\hat{f}_5|$ - 5-cycle quality;
$|\hat{f}_6|$ - 6-cycle quality.



- *One possible generalised nomenclature for the emanating harmonic qualities of the DFT coefficients.*

…diachronic summary…

…synchronic remarks…

…postlude…

…nomenclature suggestion…

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Thank you!